

KaiNinja: Extending Native 3D Generators to the Part Level

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Native 3D generators turn one image into a single mesh. TRELIS.2 and its peers deliver high-fidelity non-watertight geometry with materials, but the output is one fused object, while downstream work such as editing, rigging and simulation operates on part-level assets. A naive idea is to run a 3D segmentation network on the fused mesh that TRELIS.2 generates, but such pipelines are slow and bounded by the accuracy of the segmentation. We want a simple way to extend an existing native 3D generator to the part level. But we face a critical problem: the O-Voxel grid stores one sheet of surface per voxel, so a single volume cannot represent the interface where two parts touch, at any resolution. We introduce a dual-volume representation to solve this problem and put forward KaiNinja, a part-level extension of TRELIS.2 built on a dual-volume form of its O-Voxel representation. KaiNinja keeps the generation speed and quality of TRELIS.2 while extending it to the part level, with no mask or segmenter in the pipeline. Its training data come from sources of many kinds, including CAD models and assets authored by an LLM-driven agent; to our knowledge it is the first 3D generative model trained on agent-authored part data. Surprisingly, we also find that whole-object fidelity improves over the same backbone fine-tuned on the same dataset. Against part generation pipelines of different paradigms, it lowers whole-object Chamfer distance by 40% and raises strict part F-score by 16%.

Project page: <https://alaya-lab.github.io/KaiNinja>

Code: <https://github.com/parttrellis/PartTrellis>

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1 Introduction

Single-image 3D generation has become a practical tool recently. Native generators such as TRELIS [52], TRELIS.2 [51] and Hunyuan3D [63] denoise a 3D latent directly, and the best of them produce high-fidelity geometry with UVs, materials and PBR appearance. However, all of them return the object as one entire fused mesh, while most downstream tasks operate on parts. Editing or retexturing a component, rigging it for animation, simulating it, and reusing it in another scene all require each part to be a separate, self-contained mesh. In this paper we study *part-level image-to-3D generation*: from one image, produce an asset in which every part is its own clean mesh.

Existing methods usually obtain parts in two ways. The first way segments after generation. A 3D generator produces the whole mesh, a segmentation network such as P3-SAM [35] splits it into parts, and a part generator such as X-Part [56] regenerates every part from the mesh under the guidance of that segmentation. The second way builds part structure into the generation process itself, so that parts are generated together with the mesh from the input image. OmniPart [59] is a representative of this way. Methods in this family usually rely on part masks or predicted bounding boxes, and they generate the parts either in parallel (OmniPart) or one after another (AutoPartGen [3]). The first family and many methods in the second family depend heavily on the accuracy of a 3D mask or segmentation. When the mask is wrong, the generated parts overlap or fuse, and the generator cannot correct it. Besides, the first family also faces a serious speed problem: measured end to end, the strongest generate-then-segment cascade is an order of magnitude



Figure 1 Part-aware image-to-3D generation with KaiNinja. Each mesh is generated by our model from a single RGB image, with no 2D mask, no part segmenter, and no per-object optimization in the geometry pipeline. Materials are added with TRELIS.2 for visualization; mirror reflections use distinct colors to show individual parts. Each part is a separate, self-contained sub-mesh that can be moved, retextured, or rigged on its own.

slower than the methods of the second family (Section 4.2).

We want a fast and simple way, and therefore a more scalable one, to form parts during the generation process. Suppose we remove the part bounding boxes, which predict and constrain the number of parts, three problems then appear. The first problem is that the number of parts N is not fixed, so the model has to decide how to split the object and how many parts to generate, and this still calls for some form of segmentation or clustering. The second problem is that N can be large. AutoPartGen [3] generates parts one by one in an autoregressive manner, and PartGen [2] completes and reconstructs each part separately, so both need N passes through the network. PAct [30] instead allocates one latent volume per part and denoises them jointly, which multiplies the computation by N and caps N at a fixed maximum. We want to reduce the computation and at the same time leave the maximum N unlimited. The third problem comes from the representation of TRELIS.2 itself. Its O-Voxel grid stores one dual vertex per voxel, solved by a quadratic error function from the surface crossings of that voxel, and it extracts the mesh by dual contouring, which connects the dual vertices of neighboring voxels across each crossed edge. A single voxel therefore holds a single sheet of surface. When two nearly parallel surfaces fall into the same voxel, the quadratic error function fits one vertex between them, and the two sheets collapse into one. This is exactly the situation at a part contact, where the faces of two parts lie parallel and close.

Faced with these three problems, we notice that the dual-volume representation of PartPacker [44] solves all three at once. It packs all parts into two volumes such that touching parts never share a volume. First, parts can be read off as the connected components of each volume, so the model never predicts N and never runs a segmenter. Second, any number of parts fits into a fixed pair of streams, so the backbone runs once and N has no upper bound. Third, the contact faces of touching parts never share a voxel, so nothing collapses.

The remaining question is how to inherit the pretrained prior. The pretrained flow understands images and generates geometry from them, and we do not want to retrain these abilities. However, three gaps separate

its prior from our task. First, its prior was learned on whole objects voxelized as one volume, where the contact faces between parts collapse under the one-sheet-per-voxel rule, so it has never seen a part interface. Second, each packed volume is roughly half an object with its contact faces exposed, which is a distribution it has never seen. Third, a decomposition is a joint decision, so the two volumes must interact, and two streams that each learn their own distribution cannot agree on where one part ends and the next begins.

KaiNinja closes these gaps in order. Each stream starts from the pretrained weights and is first adapted to its own volume, so the distribution gap is crossed before any joint structure is asked of it. The two streams are then connected by new cross-volume attention that is zero-initialized, so the model begins as exactly the adapted pair and learns to coordinate from there. The two stages of the cascade take this recipe differently, because they group their latents differently: the layout flow, which decides the part structure, receives per-stream weights, while the refinement flow keeps the pretrained backbone untouched and separates the streams only through the scope of attention.

The extension keeps the speed and quality of the backbone, and it improves whole-object fidelity. At 512 generation resolution, it produces a part-separated asset in about 24 seconds per object on one H100. Surprisingly, it also generates better whole objects than the same backbone fine-tuned on the same corpus, including a 38% reduction in Chamfer distance. This suggests that packing is not only a container for parts but also a better representation of the whole objects.

We train on one corpus assembled from four 3D datasets, which include CAD models, artist-made assets and agent-authored assets. One of the four is Articraft-10K [65], a large collection of assets authored by an LLM-driven 3D creation agent. Each asset is built by a program that assembles parts from primitives, so its part labels are exact by construction. On a held-out test set spanning all four sources, KaiNinja achieves state-of-the-art performance.

In summary, our contributions are:

- The first native part-level extension of the state-of-the-art 3D object generation foundation model TRELIS.2. The recipe starts exactly at the pretrained model and treats the two stages differently, with fit, merge and warm up for the layout flow and an untouched backbone with volume-restricted attention for the refinement flow. KaiNinja demonstrates state-of-the-art performance on part-level image-to-3D generation.
- A representation-level argument for why the extension must change the representation, namely that one volume cannot hold a part interface, and its minimal remedy: dual-volume packing on the generator’s own sparse voxel grid, which keeps open surfaces, UVs, materials and PBR attributes intact.
- The first use of agent-authored 3D assets for training a 3D generative model. We train on the released Articraft-10K [65], whose assets are built by programs that assemble each part from primitives, so part labels come from authoring rather than from annotation and carry no labeling noise.

2 Related Work

Native image-to-3D generation. Most current image-to-3D methods encode shapes into a compact latent and denoise it with a diffusion or rectified flow model. 3DShape2VecSet [61] and Michelangelo [64] introduced vector-set latents over neural fields, and Dora [4] improved the VAE behind them with sharp-edge sampling. CLAY [62], CraftsMan3D [20], Direct3D [50], TripoSG [21], Hunyuan3D [63] and Hi3DGen [60] scaled this recipe to high-fidelity generation from a single image. TRELIS [52] moved the latent onto a sparse structured grid with decoders for several output formats, and TRELIS.2 [51] replaced the underlying field with the field-free O-Voxel, which supports open, non-manifold and enclosed surfaces together with PBR appearance. These models make whole objects easy to obtain, but every one of them outputs one fused geometry with no parts. KaiNinja extends this line rather than replacing it. It takes the most recent member, TRELIS.2, and generates parts on the same grid the backbone already denoises.

Multi-view and optimization-based 3D generation. Before native latents, image-to-3D methods either distilled

2D diffusion priors through score distillation (DreamFusion [37]) or generated several views (Zero123++ [40], Wonder3D [33]) and reconstructed a mesh from them (One-2-3-45 [28], InstantMesh [53]). This route is slow and often inconsistent across views, which is why later work denoises 3D latents directly. Like the native line, it returns whole objects with no part structure.

Part-aware and compositional 3D generation. Structure-aware generation predates the current wave. StructureNet [36] generates part hierarchies with graph networks, SDM-NET [10] generates deformable part meshes, SPAGHETTI [12] supports part-aware implicit edits, and SALAD [18] runs a cascaded diffusion from part layout to part geometry. These works show that parts are worth modeling explicitly, but they operate on small single-category collections rather than on open-domain images. Among current methods, the main difference is *where part structure enters the pipeline*. (i) *Segment, then regenerate*. These methods take a whole 3D object as input and split it before completing or regenerating each part. PartGen [2] segments multi-view renders and completes each part separately, HoloPart [57] completes the fragments that an external segmenter provides, and X-Part [56] runs its companion segmenter P3-SAM [35] on the object and regenerates all parts at once, guided by the segmentation boxes and features. The boundaries belong to the segmenter, and part reasoning cannot begin until a whole mesh exists. (ii) *Masks or boxes before generation*. OmniPart [59] segments the input image in 2D and lifts the masks into a bounding-box layout that steers generation, so the layout is fixed before any 3D reasoning happens. Part123 [25] reconstructs a part-aware shape from a single image with 2D masks, and ComboVerse [6] and PhyCAGE [55] segment the image into components, generate each component separately and then compose them. (iii) *Autoregressive, one part at a time*. AutoPartGen [3] generates parts in sequence, each conditioned on the parts already generated, and decides on its own when to stop. This handles a variable number of parts but costs one pass per part. (iv) *End-to-end, with no mask and no box*. These methods generate all parts from the image in one pass and differ in how they encode part identity. PartPacker [44] packs all parts into two complementary volumes that one flow denoises jointly, and PartCrafter [23] groups the latent tokens by part, with the number of parts given as input. Both are built on vecset backbones that decode a field, which requires watertight parts and models geometry alone. KaiNinja belongs to this group. It keeps the dual-volume packing of PartPacker, but moves it onto the sparse voxel grid that TRELLIS.2 already denoises, through both levels of its structured latent, and inherits the pretrained prior rather than retraining it.

A separate line reaches parts from the opposite direction. Native mesh generators model vertex connectivity and face existence explicitly. MeshGPT [41] and MeshAnything [5] tokenize faces autoregressively, and Nexus [47] replaces the sequence with diffusion over vertices and over topology, but all three generate one whole mesh. Two recent models obtain parts from the connectivity itself. LATO.2 [32] factorizes generation into a flow over vertices and a flow over topology and adds a part-wise mode, in which a structure planner partitions the scaffold with part bounding boxes, vertices are generated per part, and connectivity is predicted per part or jointly and then stitched. Meshy T2 [54] generates vertices and edges jointly with flow matching, and its vertex-set VAE keeps coincident vertices as distinct tokens instead of welding them, so touching parts are not fused and a multi-part asset comes out as connected components with no part-wise generation at all. Two things keep this line from being a comparable baseline today. First, the part-capable models are not yet available: LATO.2 publishes weights for whole meshes but not for its part-wise variant, and Meshy T2 has announced a code and weight release. Second, explicit connectivity remains hardest exactly where part structure is decided, at dense contacts and at interior surfaces the input view never shows. We treat this line as complementary.

3D part segmentation. Splitting an existing shape is the basic operation that group (i) relies on. PartSLIP [29] uses image-language models. SAMPart3D [58] and Segment Any Mesh [43] lift 2D masks from SAM [17] and SAM 2 [39] into 3D. P3-SAM [35] trains a promptable native 3D segmenter on part supervision. PartField [27] learns a feature field for grouping. When these methods run after generation, they are limited by segmentation quality and cannot recover structure the generator has already fused.

Program-driven and agentic asset generation. A separate route produces part-structured assets by writing programs rather than by denoising geometry. Infinigen [38] and Infinite Mobility [22] generate scenes and articulated objects from hand-written procedural rules, CAGE [26] generates part boxes and joint parameters from a connectivity graph and retrieves part geometry, Articulate-Anything [19] lets a vision-language model write the assembly code and retrieve parts with iterative self-correction, Artcraft [65] lets an LLM

coding agent build each part from primitives and join the parts with physical joints, and `img2threejs` [14] lets a coding agent rebuild the object in a single reference image as procedural Three.js code, with validation scripts gating each stage. Part structure, and articulation where it is modeled, are correct by construction, and the articulated assets can be simulated, but geometry is bounded by the primitives and the retrieval library. We treat this route as complementary to generative models and as a data source: the released Articraft-10K is one of our four training corpora, and its part labels come from the authoring program rather than from annotation.

Positioning. KaiNinja belongs to group (iv), together with PartPacker and PartCrafter, and it is the first method in this group built on TRELLIS.2 and its O-Voxel representation. Compared with (i) and (ii), no segmenter, no mask and no box sit on the critical path, so part boundaries are decided by the generator itself. Compared with (iii), every part is generated in one pass. Compared with the other members of (iv), the extension keeps the two-level structured latent of the backbone and its support for open surfaces and PBR materials, which a vecset/SDF representation does not offer. We compare against both X-Part cascades, OmniPart, PartPacker and AutoPartGen in Section 4.

3 Method

Overview. KaiNinja extends TRELLIS.2 to the part level, and the design follows the problems raised in Section 1. Section 3.1 changes the representation: an object with any number of parts is packed into two interleaved O-Voxel volumes, streams *A* and *B*, so that parts in the same volume do not touch. As a result, the generated geometry within each volume can be decomposed into individual parts directly by connected-component analysis, without requiring a separate segmentation network. Section 3.2 extends the backbone: the two-stage flow of TRELLIS.2 becomes a two-stream, two-stage flow that runs once for any number of parts, and each stage is adapted in the way its latent grouping asks for. Section 3.3 organizes training around the three gaps between the pretrained prior and our task, so that every phase starts from the pretrained weights and moves them as little as the task requires. Section 3.4 describes the corpus, and Section 3.5 describes inference and the two post-processing steps that turn the two volumes into parts.

3.1 The Dual-Volume O-Voxel Representation

O-Voxel. TRELLIS.2 stores geometry in O-Voxel, a sparse voxel structure built on dual contouring [16]. Each active voxel holds one dual vertex and one crossing flag per axis edge, so it stores a patch of surface rather than a sample of a field. Because no global signed distance or occupancy field is fitted, O-Voxel can represent open, non-manifold and enclosed surfaces directly, together with UVs, materials and PBR attributes. We keep all of this. An object is voxelized directly from its original textured mesh after a rigid normalization (centering, scaling into the unit cube and aligning the up axis), with no remeshing and no watertight conversion. As a result, what the backbone can represent, the extension can represent too.

One sheet per voxel. One property of O-Voxel shapes the design: a voxel holds at most one sheet of surface. Where two parts touch, their outer surfaces lie against each other as two nearly coincident sheets. At any finite resolution both sheets fall into the same voxels, and voxelization keeps one and discards the other. This is a limit of capacity rather than of resolution. A single volume that spans the whole object cannot represent a part interface, and a part interface is the structure a part-level generator has to preserve. Packing parts that touch into *different* volumes restores one sheet per voxel in each volume, so both contact faces are kept where parts meet. This is why the extension begins at the representation. We do not expect any architecture on top of a single volume to recover surfaces that the input never contained.

Packing parts into two volumes. We pack by two-coloring a contact graph *G*. Each part is a node, and two nodes are joined when the parts touch. PartPacker [44] detects contact by dilating SDF grids and measuring interpenetration. We instead read contact off the O-Voxel grid itself and make it *strict*: two parts are adjacent if and only if some voxel carries facets of both, and we weight each edge by the number of such shared voxels. When *G* is bipartite, its two color classes are the two volumes. When it is not, we contract edges greedily until it is: while an odd cycle remains, we take one, pick the edge on it with the largest weight,

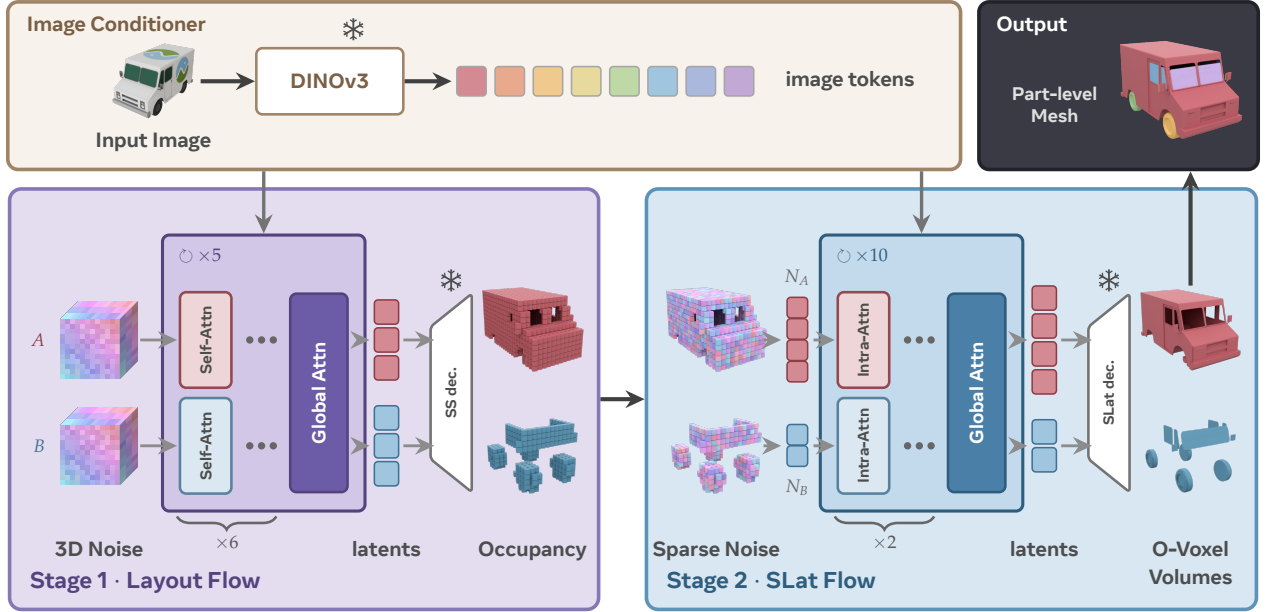


Figure 2 KaiNinja pipeline. A frozen DINOv3 encoder conditions every block of both stages through cross-attention. **Stage 1**, the layout flow, denoises two 3D noise grids into per-volume occupancy layouts. Each stream has its own self-attention weights, and a shared zero-initialized global-attention block connects the streams every six blocks ($\times 5$). **Stage 2**, the refinement flow, keeps the TRELLIS.2 weights and denoises sparse noise on the active voxels (N_A , N_B tokens) with two intra-volume blocks and one global block per group ($\times 10$). Frozen VAEs decode the latents of both stages, and the two O-Voxel volumes are assembled into the part-separated mesh. No mask or segmenter appears in the pipeline.

merge its two parts into one node, and rebuild the graph. The result is then two-colored by breadth-first search. This is the greedy odd cycle contraction of PartPacker (their Algorithm 1), a heuristic for the bipartite contraction problem [11]; it makes no optimality claim and merges the parts that are in closest contact first. In the end every part belongs to stream A or B , and parts within a stream do not touch by construction (Figure 3).

Two-coloring is a heuristic rather than a guarantee, since it works only when the contact graph is bipartite, and objects whose parts interlock densely are merged more than their annotation intends. PartPacker suggests, as a possible remedy, converting the contact graph into a planar graph and applying the four color theorem. We believe this suggestion does not hold in general: a 3D contact graph need not be planar, and making it planar deletes contact edges, which puts touching parts back into one volume. Appendix B gives the argument and explains why no fixed number of volumes removes this limit.

3.2 Two Streams, Two Stages

One recipe per stage. TRELLIS.2 generates in a cascade of two rectified flows, and the two stages group their latents differently. The first stage denoises a coarse structure latent that decides *where* surface exists. The second stage denoises a structured latent that decides *what* the surface is at each occupied voxel. Part structure lives mostly in the first decision, because which parts exist, where they sit and which volume each belongs to is a joint allocation over both streams. The second stage, in contrast, is local refinement that the pretrained prior already performs well. For this reason we treat the two stages differently. The layout flow gets per-stream weights connected by new cross-volume blocks, while the refinement flow keeps the pretrained backbone unchanged and separates the streams only through the scope of attention. Both choices are inheritance decisions, and we explain each below.

Conditioning. A DINOv3 ViT-L/16 encoder [42] reads the image at 512×512 and produces conditioning tokens. These tokens enter both stages through cross-attention, as in the backbone. We use classifier-free

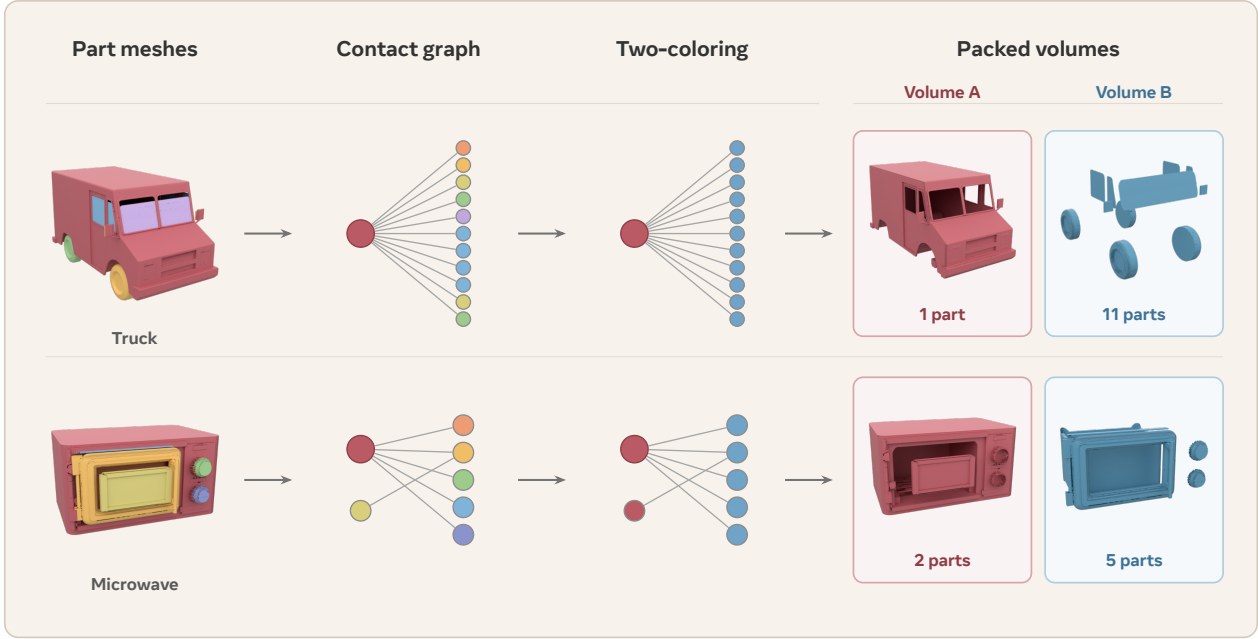


Figure 3 Dual-volume packing. For each object we build its part-contact graph (nodes keep the part colors of the mesh), two-color it into the two streams, and pack each color class into one O-Voxel volume. The truck is a star (body against wheels, windows and mirrors); the microwave splits into 2+5 parts with six contacts. Parts that share a volume do not touch by construction, so each volume can be decoded and split into parts on its own.

guidance [13]: one denoising pass with the image and one with a null condition, extrapolating past the conditional prediction. Nothing in either stage is specific to images, so other conditioning signals could be used instead.

Stage 1: the layout flow. The first stage denoises the coarse occupancy of both volumes together. It works on the sparse structure latent, a 16^3 grid with 8 channels per stream, which the pretrained decoder expands to a 64^3 occupancy grid. Each stream is a full copy of the pretrained transformer (30 blocks, width 1536, 12 heads, rotary position encoding), with its own input and output projections and its own timestep modulation. Stream identity is therefore carried by the weights and needs no extra embedding. After the per-stream blocks at depths $\{6, 12, 18, 24, 29\}$, one shared *cross-volume attention* block attends over the tokens of both streams at once, with a modulation branch of its own. These blocks are where the model reasons jointly, for example when it decides where two touching parts should split. They are zero-initialized, so at the start of training the network is two independent pretrained denoisers, each seeing one volume. The training schedule of Section 3.3 relies on this property.

Stage 2: the refinement flow. Given the predicted layout and the same image tokens, the second flow denoises the structured latent of both streams, which carries fine geometry and appearance. Occupancy is fixed by now, so the task is close to the one the backbone was pretrained for, and we leave the backbone unchanged. Its single weight set is loaded as released, and the two streams coexist by *scoping attention*. In 20 of the 30 blocks each self-attention layer attends only over the tokens of one volume, so each stream refines on its own. In the remaining 10 blocks (2, 5, 8, $\dots$, 29) it attends over both volumes at once. Self-attention handles a variable token count natively, so this regrouping changes no layer width and adds no backbone parameters. In the merged blocks, stream identity comes from a learned volume embedding, which is initialized at zero and added to the timestep modulation.

3.3 Training: Inheriting a Whole-Object Prior

The prior we inherit was trained on whole objects. It understands images and generates geometry from them, and these are the abilities we want to keep. However, three gaps separate it from our task (Section 1):

its training data were voxelized as whole objects, so it has never seen a part interface; a packed volume is roughly half an object with its contact faces exposed, which is a distribution it has never seen; and the two volumes have to interact. We therefore organize training so that each phase moves the weights the least distance the task requires. Every phase uses a flow matching objective [24, 31].

The layout flow: fit, merge, warm up, fine-tune. Stage 1 is trained in three phases. (i) *Fit each volume.* Starting from the pretrained single-stream denoiser, we fine-tune two copies independently, one on the volume- A latents of our corpus and one on the volume- B latents. This step carries the prior across the distribution gap. Each copy adapts to the marginal distribution of its own stream, that is, half objects with exposed contacts, before any joint structure is asked of it. (ii) *Merge without loss.* The two fitted copies are loaded as the two streams. Every component they touch (blocks, projections and timestep modulation) belongs to one stream, so the fitted weights transfer without reinterpretation. Together with the zero-initialized cross-volume blocks, the merged model starts out behaving like the two fitted denoisers. (iii) *Warm up, then fine-tune jointly.* A frozen warmup holds both streams fixed and trains only the cross-volume attention and its modulation branch, so the streams learn to coordinate before their fitted priors are disturbed. Joint fine-tuning then unfreezes everything. This order is meant to protect the inheritance. Skipping the warmup would let untrained cross-volume blocks push gradients through fitted streams, and skipping the per-volume fit would ask one set of pretrained weights to serve two marginals it has never seen.

Throughout joint fine-tuning we add a *disjointness penalty* to the flow matching loss. It targets the layer where part structure is decided. Packing puts every part in exactly one volume, so the two streams should not claim the same voxel, but nothing in a per-stream flow loss says so. At each step we recover the predicted clean latent $\hat{x}_0$ from the velocity, decode both streams with the frozen sparse structure decoder into soft occupancies $o_A, o_B \in [0, 1]$ on the 64^3 grid, and penalize the overlap, $\mathcal{L}_{ov} = \mathbb{E}[o_A \odot o_B]$, added to the flow loss with weight $\lambda_{ov} = 5$. It costs one decoder pass per step, and we examine its effect in Section 4.5.

The refinement flow: warm up, fine-tune. Stage 2 needs no per-volume fitting. Its backbone is unchanged and both streams reuse the pretrained weights as released, so there is no distribution gap for the weights to cross before joint training. It trains on the same two-phase schedule: a frozen warmup that updates only the volume embedding, followed by full fine-tuning. In our experience this is enough, because scoping attention disturbs the pretrained computation far less than changing the stream topology would. The two stages therefore receive different treatment, which is what inheriting one whole-object prior into a cascade seems to require when the stages group their latents differently.

3.4 Data Curation

We assemble one corpus from four 3D datasets: a large collection authored by a 3D creation agent rather than by human artists (ARTICRAFT [65]), a CAD corpus (FUSION360 [49]), and two collections made by artists (PARTNEXT [48] and TRELLIS [52]). Artcraft turns the creation of an articulated asset into code generation driven by a language model. The agent writes a program that builds each semantic part out of primitives and joins the parts with physical joints. Its parts are therefore not annotations but the structure the agent wrote, so the part supervision it provides is exact. We use the released Artcraft-10K [65] collection and pass it through the same filters as the other sources. To our knowledge this is the first time assets authored by an agent have been used to train a 3D generative model. For each object we read part annotations off the mesh scene graph, falling back to connected components when one geometry node fuses everything. We then voxelize the raw textured mesh into O-Voxel and run the merge-and-two-color procedure of Section 3.1 to obtain the packing.

Quality filtering. The corpus passes two filters. Before voxelization, on the normalized meshes, we keep objects with 2 to 32 parts, drop objects whose parts interpenetrate badly, measured both by the depth of pairwise collisions and by the overlap between parts under a generalized winding number [15], and remove assets dominated by display planes baked in as ground or backdrop quads. After packing, we require a valid latent and a rendered condition for every object. We also apply a balance filter, a minimum voxel count and a minimum occupancy ratio between A and B , which discards degenerate packings that leave one volume empty or nearly so. PartPacker applies no such check. Because O-Voxel is field-free, we skip

the watertight repair and shell dilation that packing over an SDF [44] requires, and we keep the original open surfaces and materials.

3.5 Inference and Post-processing

At inference the layout flow samples the structure latents of both volumes from the image, and the refinement flow fills them in. Two light post-processing steps then turn the two volumes into parts, and neither involves a segmenter or a mask.

Merging duplicated occupancy. Nothing forces the two streams to stay disjoint at inference, and they occasionally place the same surface in both volumes. Between the stages we therefore compare the decoded occupancies. For every pair of connected components, one from each volume, we measure the fraction of shared voxels and the depth of embedding. When a pair overlaps by more than half and is embedded at least three voxels deep, the voxels of the smaller component move into the volume of the larger. Nothing is deleted, so the union of the two volumes is unchanged. The step adds negligible cost. Thin shells sometimes escape it (Section 5).

Parts from volumes. The frozen TRELLIS.2 VAE decodes the predicted latent of each volume into an O-Voxel volume, and a textured mesh is extracted from each. Within a volume, parts do not touch by construction, so we take the connected components of each mesh as parts. Because the decoder sometimes leaves small gaps, we treat components whose surfaces lie within two fine voxels of each other as one part, and attach fragments below 0.5% of the volume’s surface area to the nearest part. The two meshes are then assembled into the finished object, followed by a light geometric cleanup. Neither step predicts the number of parts; it follows from the layout the model generated. Section 4.5 measures what each step contributes.

4 Experiments

4.1 Setup

Data. We train on one corpus of 19,132 objects drawn from four 3D datasets (ARTICRAFT [65] 8,968, PARTNEXT [48] 6,358, TRELLIS [52] 1,931, FUSION360 [49] 1,875). Each object is packed into two O-Voxel volumes as in Section 3.4. We treat the four sources as one corpus rather than as separate domains: every object goes through the same pipeline, and the flows are trained jointly on the union. We hold 1000 objects out of training as a test set that spans all four sources (ARTICRAFT 513, PARTNEXT 297, FUSION360 102, TRELLIS 88), and we report a breakdown by source. One baseline, OmniPart, produces no output on 14 of the 1000 objects because its sparse convolution backend runs out of resources on very dense voxel grids. To keep every method on identical inputs, we report all methods on the 986 objects that every method completes.

Baselines. We compare against a representative of each family in Section 2: X-Part [56], which segments a generated mesh and regenerates its parts; OmniPart [59], which plans a layout from masks; AutoPartGen [3], which generates one part after another; and PartPacker [44], which packs dual volumes over a vecset latent and is the method closest to ours. X-Part takes an existing mesh rather than an image, so it needs a generator in front of it. We use TRELLIS.2 [51], the backbone our own flows extend, at its released default 1024_cascade setting, and write this cascade as TRELLIS.2@1024 + X-Part. The X-Part paper feeds it Hunyuan3D-2.5, whose weights are not released, so we also report it on Hunyuan3D-2.1 [45], the released model closest to its own setup. The generator in front matters: moving X-Part from TRELLIS.2 to Hunyuan3D-2.1 raises it by $0.038 F1_p^{0.05}$, and we report both pairings throughout. Neither choice handicaps X-Part, since our own Stage 2 starts from the 512 shape flow of TRELLIS.2 and never runs the 1024 cascade. Every method therefore starts from the same single image and is scored by the same protocol. Each baseline runs its own released pipeline, with mesh simplification and decimation switched off so that geometry is scored as the model decodes it.

Metrics. Following OmniPart [59], we normalize every shape into $[-0.5, 0.5]^3$ and report Chamfer distance (CD) and F-score at thresholds 0.1 and 0.05, for the whole object (W) and for its parts (P). Methods use different canonical frames, so for each object we align the prediction to the ground truth by picking, out of

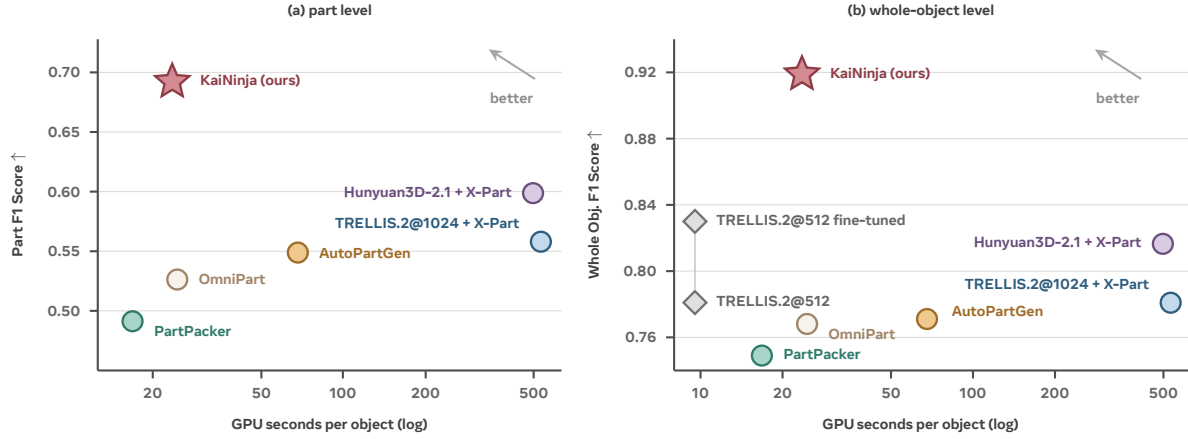


Figure 4 Quality against generation cost, at (a) the part level and (b) the whole-object level. Both vertical axes are F-scores at the strict 0.05 threshold ($F1_p^{0.05}$ and $F1_w^{0.05}$ of Table 1). The horizontal axis is GPU seconds per object on a logarithmic scale, so up and to the left is better. Gray diamonds in (b) are two whole-object references of Table 2: TRELLIS.2@512 zero-shot, and the same model with both flow denoisers fine-tuned on our corpus. Fine-tuning leaves the architecture unchanged, so the pair shares one measured cost. They emit one undivided mesh, so they have no point in (a), and the x axis of (a) therefore starts later. KaiNinja sits in the upper left of both panels: close to PartPacker in cost, and above every method in quality.

48 signed axis permutations, the one with the smallest CD on the whole object, and we reuse that transform when scoring parts. Each method contributes the set of parts its own pipeline produces, and we impose no further splitting or filtering on any method. For KaiNinja a part is a connected component of a decoded volume (Section 3.5). Methods that emit parts directly contribute those parts as released. Both routes end at the same kind of object, a set of parts whose size is not tied to the ground truth, which is what the metric consumes.

Ground-truth and predicted parts are matched *one-to-one* by Hungarian assignment, at cost $1 - \text{IoU}$ on surface labels transferred by nearest neighbor. We match on IoU rather than CD because IoU is bounded, so one badly placed part cannot dominate the assignment. Predictions left unmatched and ground-truth parts left uncovered both count against the method, so over- and under-segmentation are both penalized. On the matched pairs we report the mean IoU ($m\text{IoU}_p$) and CD / F-score. OmniPart’s paper does not specify how predicted parts are matched to ground truth, and no compared method releases part-metric code, so the protocol above is our own choice.

Implementation. DINOv3 ViT-L/16 [42] encodes the input image at 512×512 . Both flows use the two-stream configuration of Section 3.2, are trained on the frozen-warmup then joint-fine-tuning schedule, and run with classifier-free guidance at inference. The structured latent autoencoder is the pretrained TRELLIS.2 one and stays frozen throughout. We train only the two flows, so Stage 2 operates in exactly the released latent space and no autoencoder is retrained. Appendix A lists the hyperparameters.

4.2 Comparison with State of the Art

Part-level quality. Table 1 gives the overall comparison and Table 3 the breakdown by source, both on the 986 objects every method completes. KaiNinja is best on all seven metrics of Table 1 and on every entry of the breakdown. Against the strongest baseline, Hunyuan3D-2.1 + X-Part, it gains $0.10 F1_p^{0.05}$ and $0.09 F1_w^{0.05}$ at the strict threshold, and its whole-object Chamfer distance is 40% lower. The gap is small at the loose threshold and widest at the strict one, which suggests that the difference lies in fine geometry and boundary placement rather than in coarse layout.

Cost. Figure 4 reports the measured cost of every method. Cost is wall-clock time on H100s over the same 986 objects, converted to GPU seconds. We run one process per GPU for every method except X-Part, whose segmenter occupies a whole four-GPU node, so its 117s wall clock is charged as 469 GPU seconds. The

Table 1 Part-aware image-to-3D. Whole-object (W) and Hungarian-matched part (P) metrics of Section 4.1, on the 986 held-out objects all six methods complete.

Method	$CD_W \downarrow$	$F1_W^{0.1}$	$F1_W^{0.05}$	$mIoU_P$	$CD_P \downarrow$	$F1_P^{0.1}$	$F1_P^{0.05}$
TRELLIS.2@1024 [51] + X-Part [56]	0.0350	0.900	0.780	0.449	0.0804	0.703	0.559
Hunyuan3D-2.1 [45] + X-Part [56]	0.0314	0.917	0.814	0.489	0.0764	0.728	0.597
OmniPart [59]	0.0366	0.901	0.769	0.456	0.0913	0.664	0.518
PartPacker [44]	0.0389	0.892	0.749	0.449	0.0986	0.640	0.492
AutoPartGen [3]	0.0373	0.901	0.771	0.466	0.0919	0.678	0.535
KaiNinja (ours)	0.0186	0.975	0.919	0.521	0.0613	0.785	0.692

measurement covers the whole pipeline, generation and mesh export together. For the 1024 cascade the export dominates: 51.2 of its 62.4s go to writing and decimating a mesh of 692K vertices, against 11.1s of generation. Both X-Part cascades pay the segmenter charge and differ only in the generator in front, so they sit close together on the axis. PartPacker is the cheapest method and the weakest on parts; our model costs 40% more and gains 0.20 $F1_P^{0.05}$. AutoPartGen spends $3\times$ our budget, because it generates one part after another, and the segment-then-regenerate cascade costs an order of magnitude more than any other method, because it cannot begin part reasoning until a whole object exists. The panels also show that whole-object quality separates these methods very little: four of the five baselines sit within 0.032 of one another, and the fifth is 0.035 above the best of them. The methods separate once parts are scored.

Whole-object quality. Table 2 shows what the extension costs and what it improves. Four of the five part-producing baselines score at or below every zero-shot whole-object reference at the strict threshold, 0.749 to 0.781 against 0.781 to 0.807. In other words, reaching parts from outside the generator usually costs fidelity. Only the Hunyuan3D-2.1 cascade is higher, at 0.816, and only 0.009 above its own upstream. The fine-tuned row then separates what the corpus contributes from what the extension does. Giving TRELLIS.2@512 the same fine-tuning data raises it to 0.830 and cuts its failure rate to 1.8%. KaiNinja starts from the same pretrained weights and sees the same corpus, yet it is another 0.089 above that at about $2.5\times$ the generation cost (Figure 4b), and it delivers parts. At 0.919 it is above every reference in the table, and it misses the shape outright on 0.7% of objects, against 3.9 to 6.8% for the zero-shot references. We did not expect this gain, and we read it as evidence that packing is a better representation of the same objects rather than only a container for parts. Two details qualify this reading. Running TRELLIS.2 as a 1024 cascade does not help on this corpus (0.0364 against 0.0363). Also, TRELLIS and Hunyuan3D-2.1 share a mean but not a distribution: Hunyuan3D-2.1 has the better median CD_W , 0.0209 against 0.0235 (not shown in the table), and nearly twice the failure rate.

X-Part and its generator. X-Part’s score tracks the generator in front of it. Paired with Hunyuan3D-2.1 it reaches 0.816 $F1_W^{0.05}$, within 0.009 of what Hunyuan3D-2.1 reaches alone (Table 2), and the same holds on TRELLIS.2. Segmenting after the fact changes the whole-object score very little, and it improves the part score about as little, so the cascade scores roughly what its generator hands it. To see *which* property is inherited, we reran the cascade from the 512 shape flow of TRELLIS.2 instead of the 1024 cascade, paired over the 985 objects X-Part completes from both.¹ The part scores came out indistinguishable: the per-object differences scatter far more widely than the gap between the two means. Swapping the *model* in front, by contrast, moves X-Part by 0.035 $F1_W^{0.05}$. The cascade therefore appears to inherit the shape of the upstream generator rather than its compute budget, and it cannot exceed the mesh it is handed.

The role of the representation. PartPacker shares our packing principle and is likewise free of masks and segmenters, so the main difference between the two is where the volumes live: in a vecset latent decoded as an SDF field there, and in the backbone’s own sparse voxel grid here. On the whole object the gap is moderate (CD_W 0.0390 against 0.0186), but on parts it is the widest in the table: $F1_P^{0.05}$ 0.492 against 0.692, with CD_P more than 60% larger. The SDF representation also requires watertight preprocessing and drops the materials. We take this as support for building the packing on the backbone it extends, with the caveat that the two systems also differ in training corpus, so this comparison does not isolate the representation on its own.

¹One FUSION360 object crashes X-Part on the 512 mesh but not on the 1024 one, and is dropped from this paired comparison only.

Table 2 Whole-object generators, for reference. One undivided mesh each, so only the whole-object columns are defined. *Fail* is the fraction of objects with $CD_W > 0.1$.

Method	$CD_W \downarrow$	$F1_W^{0.1}$	$F1_W^{0.05}$	Fail % $\downarrow$
TRELLIS [52]	0.0340	0.916	0.807	3.9
TRELLIS.2@512 [51]	0.0363	0.897	0.781	5.3
TRELLIS.2@1024 [51]	0.0364	0.898	0.781	5.4
TRELLIS.2@512, fine-tuned on our corpus	0.0300	0.940	0.830	1.8
Hunyuan3D-2.1 [45]	0.0340	0.912	0.807	6.8
KaiNinja (ours, part-aware)	0.0186	0.975	0.919	0.7

Table 3 Breakdown by source (whole-object $F1_W^{0.1}$ / Hungarian-matched part $F1_P^{0.05}$) on the 986 objects every method completes (ARTICRAFT 499, PARTNEXT 297, FUSION360 102, TRELLIS 88). Best in **bold**.

Method	ARTICRAFT		PARTNEXT		FUSION360		TRELLIS	
	$F1_W$	$F1_P$	$F1_W$	$F1_P$	$F1_W$	$F1_P$	$F1_W$	$F1_P$
TRELLIS.2@1024 [51] + X-Part [56]	0.911	0.559	0.898	0.584	0.851	0.512	0.899	0.529
Hunyuan3D-2.1 [45] + X-Part [56]	0.928	0.614	0.922	0.612	0.866	0.548	0.901	0.509
OmniPart [59]	0.913	0.519	0.896	0.536	0.844	0.470	0.917	0.499
PartPacker [44]	0.913	0.517	0.874	0.466	0.835	0.463	0.899	0.466
AutoPartGen [3]	0.914	0.544	0.901	0.538	0.828	0.481	0.912	0.531
KaiNinja (ours)	0.985	0.738	0.967	0.666	0.956	0.627	0.964	0.595

4.3 Generalization Beyond the Fine-Tuning Corpus

To test generalization beyond our curated splits, we also evaluate on objects from Sketchfab and GitHub, filtered down to a single connected component. We score the 95 of them that every method completes, so every number in the table comes from the same objects. Neither source feeds our fine-tuning corpus or our test split. That separation is what this set is designed to test, and it holds for every method compared. We do not call these objects unseen, because the pretrained TRELLIS.2 weights our flows start from were trained on data derived from Objaverse [8, 7], which draws on both sources, and the same holds for the generators in front of the baselines. Every method receives the same rendered view. Table 4 gives the comparison. KaiNinja keeps a clear lead on the whole object, by $0.075 F1_W^{0.05}$ over the next method. On parts it is ahead on Chamfer distance and on the strict $F1_P^{0.05}$, where the margin over the closest baseline is 0.036. The two X-Part cascades stay with us only at the loose threshold, $F1_P^{0.1}$ 0.862 against our 0.861, and both carry a higher $mIoU_P$ (0.644 and 0.624 against our 0.619). A dedicated segmenter seems to place boundaries a little more conservatively on irregular inputs, which helps once a coarse tolerance is allowed. The ordering at the strict threshold is the one that matters for downstream use, and there a model with no segmenter is ahead.

An external benchmark. The set above still shares its sources with the pretraining data. HY3D-Bench does not. It is an outside benchmark with its own part annotations, and no method here was trained on it. Every method reads the same conditioning image, one of the views HY3D-Bench itself publishes, and we score all 200 annotated objects. Table 5 reports the result, and the result splits.

On the whole object KaiNinja is first on all three metrics, by 2.6 points of $F1_W^{0.05}$ over the next method. On parts it is not. Hunyuan3D-2.1 + X-Part leads every part metric, and on $mIoU_P$ we sit fourth of six, behind AutoPartGen and PartPacker as well. This reverses the ordering of Table 1, and we attribute it to granularity rather than geometry. HY3D-Bench annotates parts more coarsely than our corpus does. Our model is trained to separate at contacts, so it cuts a chair into more pieces than the benchmark labels, and Hungarian matching charges for every extra piece. A pipeline that segments after generating a whole object inherits its granularity from the segmenter, which is easier to tune toward whatever convention a benchmark uses. We report the split as it is: on this benchmark our geometry is the most faithful, and our part decomposition is finer than the labels reward.

Table 4 Sources outside the fine-tuning corpus: the 95 Sketchfab and GitHub assets, single connected component, that every method completes. Same method set and same row order as Table 1.

Method	$CD_W \downarrow$	$F1_W^{0.1}$	$F1_W^{0.05}$	$mIoU_P$	$CD_P \downarrow$	$F1_P^{0.1}$	$F1_P^{0.05}$
TRELLIS.2@1024 [51] + X-Part [56]	0.0277	0.939	0.826	0.624	0.0459	0.862	0.717
Hunyuan3D-2.1 [45] + X-Part [56]	0.0274	0.937	0.826	0.644	0.0507	0.835	0.706
OmniPart [59]	0.0318	0.928	0.804	0.569	0.0579	0.815	0.659
PartPacker [44]	0.0317	0.926	0.791	0.607	0.0614	0.792	0.651
AutoPartGen [3]	0.0306	0.942	0.812	0.614	0.0543	0.833	0.678
KaiNinja (ours)	0.0195	0.969	0.901	0.619	0.0444	0.861	0.753

Table 5 HY3D-Bench, 200 part-annotated objects, every method reading the same released view. Shading marks first, second and third per column.

Method	$CD_W \downarrow$	$F1_W^{0.1}$	$F1_W^{0.05}$	$mIoU_P$	$CD_P \downarrow$	$F1_P^{0.1}$	$F1_P^{0.05}$
TRELLIS.2@1024 [51] + X-Part [56]	0.0516	0.807	0.639	0.344	0.0852	0.666	0.476
Hunyuan3D-2.1 [45] + X-Part [56]	0.0441	0.844	0.699	0.412	0.0728	0.729	0.555
OmniPart [59]	0.0475	0.834	0.685	0.354	0.0897	0.665	0.490
PartPacker [44]	0.0566	0.788	0.632	0.381	0.0904	0.657	0.476
AutoPartGen [3]	0.0512	0.821	0.666	0.399	0.0905	0.665	0.498
KaiNinja (ours)	0.0413	0.865	0.724	0.356	0.0787	0.688	0.505

4.4 Qualitative Results

Figures 5, 6 and 7 compare separated parts against both X-Part cascades, OmniPart and PartPacker on three groups of objects: the held-out test set, the training corpus, and objects whose source is not used in training at all. Every prediction is rotated into the ground truth frame by its best axis permutation, the same alignment the metrics use, so orientations are directly comparable, and colors mark parts rather than materials. KaiNinja gives clean and coherent parts on rigid CAD shapes, on articulated furniture and on organic objects. It also keeps the part count close to the ground truth, whereas the baselines tend to fuse parts, as X-Part does on the drinks can, or to shatter them, as OmniPart does on the bundle of pipes.

4.5 Ablation

Three choices decide how parts come out of this model: how many volumes the packing uses, whether the layout flow is penalized for putting both volumes in the same place, and what the post-processing does with the result. We take them in that order.

Number of volumes. The obvious generalization is to pack into three volumes. The contact graph is then three-colored greedily instead of contracted until bipartite, and the flow runs three streams (Figure 8b). Backbone, data and recipe are identical, so the only thing that changes is the number of volumes. Table 6 compares them at the end of Stage 1, where the designs differ. A prediction is the set of occupied voxel centers at 64^3 . Ground truth is 100K points from the continuous surface of the whole object and 20K from each part, and parts are matched by Hungarian assignment on voxel-set IoU, with no merge and no relabel. $F1^{0.1}$ saturates in this setting because one voxel spans 0.016, so a 0.1 tolerance is over six voxels wide, and we read the 0.05 column instead.

Three volumes are not worse at geometry. The whole object ties, and on parts three volumes are slightly ahead, by 1.6 points of $F1_P^{0.05}$ and 0.0067 of CD_P , both significant under a paired bootstrap. The part counts suggest why. Three volumes leave 4.73 connected components per object against 3.84 for two, with 5.05 in the ground truth, so a third volume gives touching parts one more way to avoid being fused. What three volumes cost is robustness and supervision. They fail outright on 13 of 1000 objects where two volumes never do, and they leave a stream empty on 7.7% of objects against 1.5%, so a third of the capacity idles on one object in thirteen while every step costs half again as much. We choose two volumes for robustness, not for a fidelity advantage.

The disjointness penalty. Table 7 crosses the overlap term $\mathcal{L}_{ov}$ of Section 3.3 with the three levels of post-

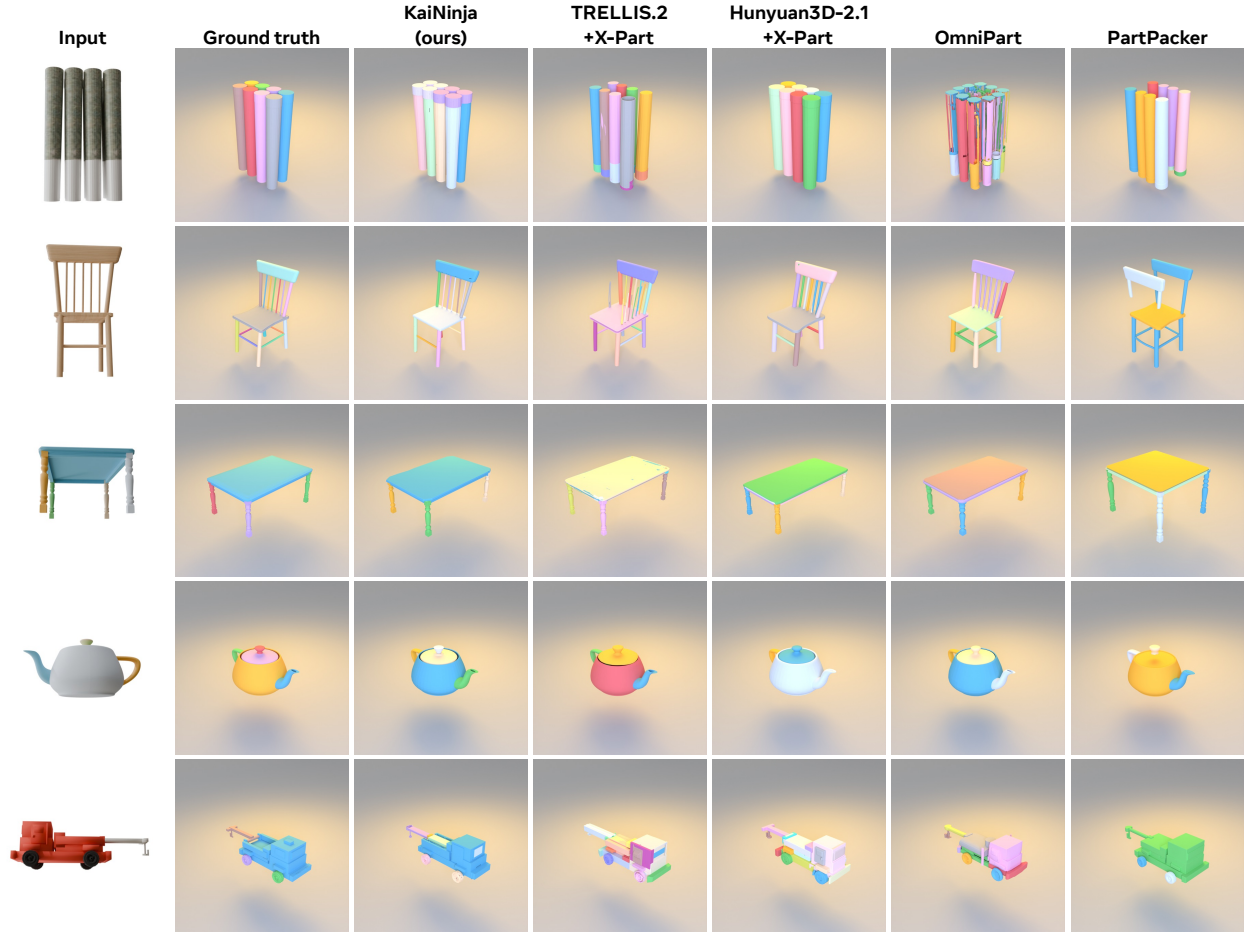


Figure 5 Held-out test set. Columns: the input image, the ground truth parts, and the five methods. Colors mark parts, not materials. Each method uses its own canonical frame, so every shape is rotated into the pose of the ground truth cell before rendering, and all cells then share one camera, one light and one scale. KaiNinja keeps the part count close to the ground truth where the baselines fuse or shatter parts. SPACE PROMPT

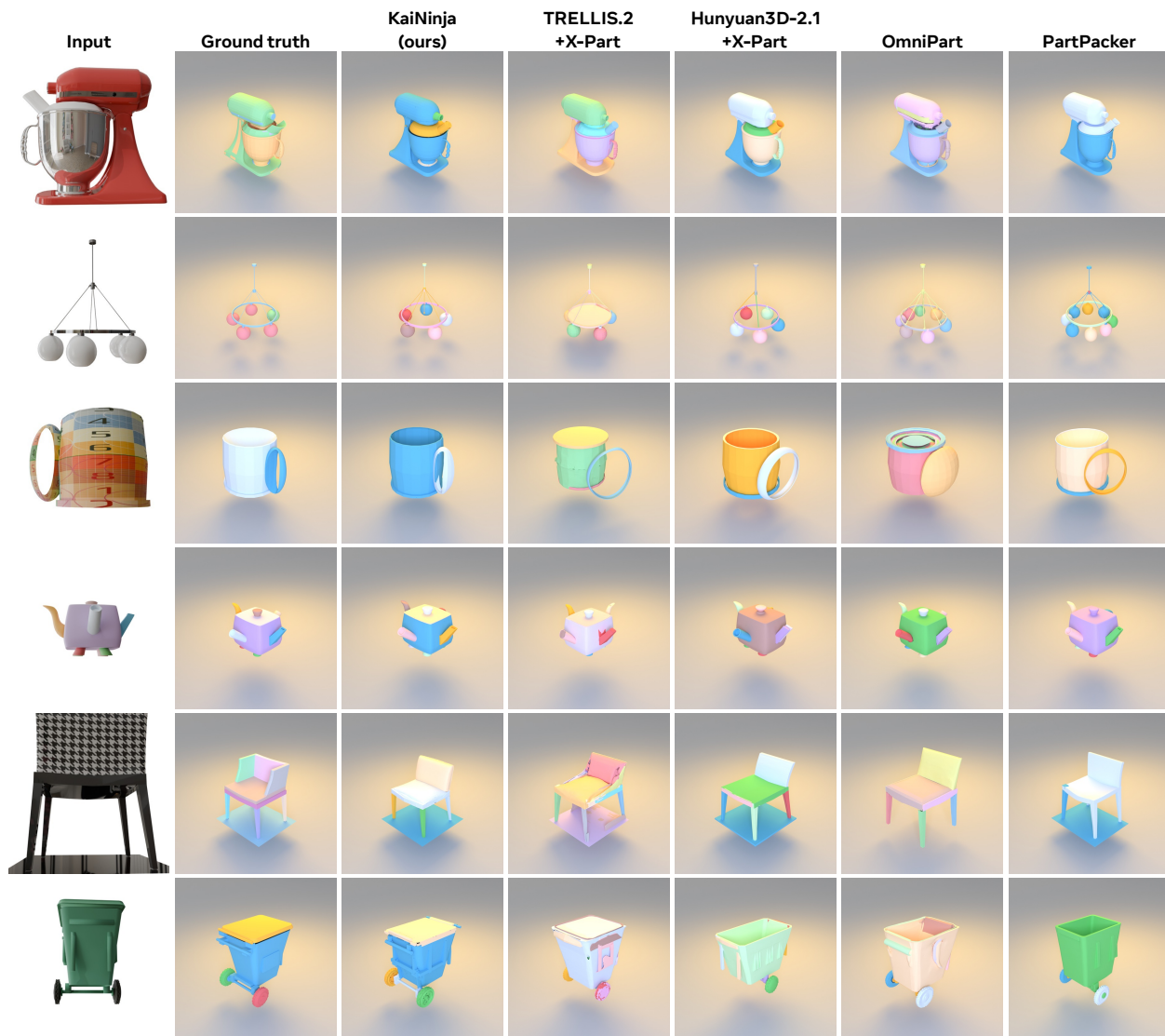


Figure 6 Training corpus. Same columns, coloring and alignment as Figure 5.

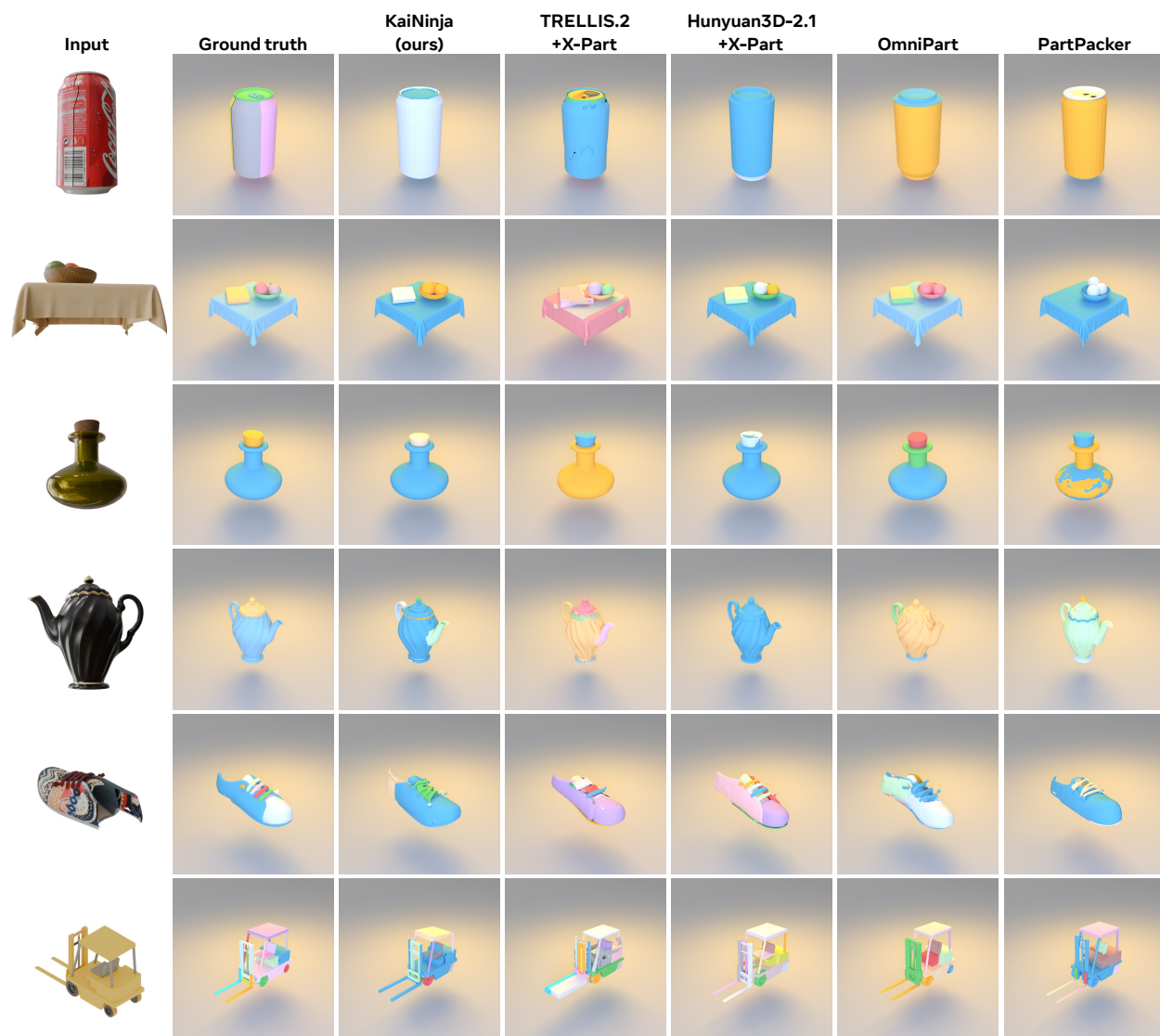


Figure 7 Sources not used in training. Same columns, coloring and alignment as Figure 5.

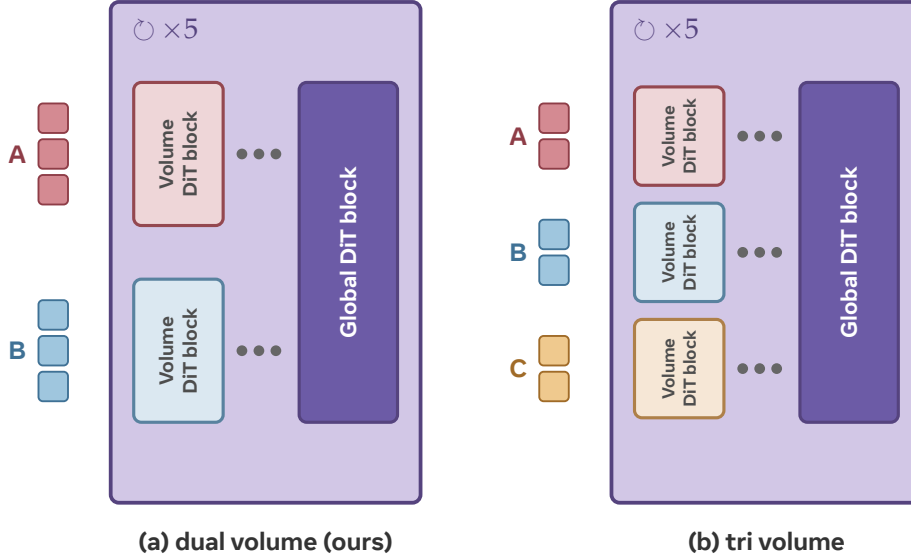


Figure 8 Ablated Stage-1 variants. Both variants run the same block budget: each repeat applies six *Volume DiT blocks* per stream and one *Global DiT block* that pools the tokens of all streams and attends across them, for 30 and 5 blocks in total. They differ only in how many volumes the parts are packed into: (a) two volumes, from the bipartite packing of Section 3.1; (b) three volumes, from a greedy three-coloring of the part-contact graph. Table 6 compares them.

Table 6 Two volumes against three, scored on Stage-1 occupancy alone over the 986 objects both arms complete. Three volumes additionally fail on 13, two volumes on none.

Variant	$CD_W \downarrow$	$F1_W^{0.1}$	$F1_W^{0.05}$	$mIoU_P$	$CD_P \downarrow$	$F1_P^{0.1}$	$F1_P^{0.05}$
Tri-volume (3 streams)	0.0428	0.961	0.908	0.306	0.0834	0.875	0.795
Dual-volume (ours)	0.0423	0.963	0.908	0.301	0.0901	0.861	0.779

processing. On the decoded output the two arms are indistinguishable. At every level they agree to within 0.006 on each metric, with the arm trained without the term consistently and slightly ahead, and the component counts agree to within 0.1 of a part. The effect of the term appears one stage earlier, where it is aimed. Over 250 held-out objects scored at the Stage-1 occupancy, the term lowers the median cross-volume overlap from 0.229 to 0.184, and it lowers the fraction of objects in which one volume largely duplicates the other, overlap above 0.5, from 17.2% to 14.4%. Figure 9 shows three objects under both arms. Without the term, volume *B* repeats surfaces that already exist in volume *A*, such as the whole bundle of pipes or the rims of the mug; with the term, *B* carries its own parts. The effect is not uniform: across the objects of the qualitative figures the term lowers the overlap on most, leaves some unchanged, and raises it on a few. We keep $\mathcal{L}_{ov}$ in the shipped model because it acts on the layout, the stage that decides part structure, and it costs one decoder pass per step.

Post-processing. The same table isolates the two post-processing steps of Section 3.5 along its first column. The merge changes the metrics very little; it is a safeguard against duplicated volumes, which the trained model sometimes produces. But the relabel matters as the final step, which attaches floating facets back onto their parts and joins the pieces of one part. This trick cuts the component count from 14.06 to 5.43 against 5.85 in the ground truth and improves $mIoU_P$ and $F1_P^{0.05}$.

5 Conclusion

We presented KaiNinja, a part-level extension of a native 3D generator. TRELLIS.2 is extended in place: its O-Voxel representation is packed into two interleaved volumes, its cascade is adapted stage by stage, and its VAE and material pipeline stay untouched. The result turns one image into an asset whose parts are separate meshes, at a small constant increase in generation cost and with no mask or segmenter anywhere

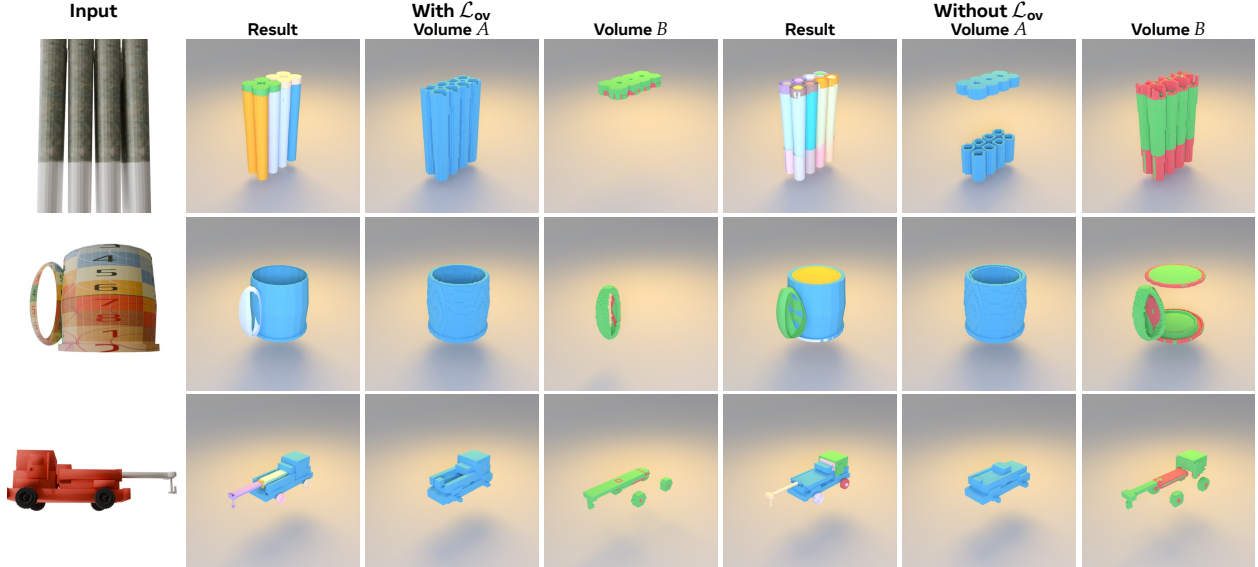


Figure 9 Effect of the disjointness penalty. Three objects from the qualitative figures, generated with and without $\mathcal{L}_{ov}$ from the same image and seed. For each arm we show the final parts and the Stage-1 occupancy of the two volumes. Volume A is blue; in the B columns, green marks voxels that only B occupies and red marks voxels claimed by both volumes. Without the term, volume B repeats surfaces that already exist in A : the whole bundle of pipes, the rims of the mug, the chassis of the truck. With the term, B carries its own parts. The overlap $|A \cap B| / \min(|A|, |B|)$ drops from 0.64 to 0.31, from 0.21 to 0.14 and from 0.22 to 0.09. These objects were chosen for a visible difference; on many others the two arms are close, and on a few the term does not help. After post-processing the final parts differ little in either case, which matches Table 7.

Table 7 Post-processing and the disjointness penalty, on the 977 test objects all six arms complete. Ground truth averages 5.85 parts.

Post-processing	$\mathcal{L}_{ov}$	$CD_W \downarrow$	$F1_W^{0.1}$	$F1_W^{0.05}$	$mIoU_P$	$CD_P \downarrow$	$F1_P^{0.1}$	$F1_P^{0.05}$	#parts
none	✓	0.0195	0.973	0.913	0.481	0.0627	0.783	0.675	14.06
none	✗	0.0191	0.975	0.917	0.482	0.0621	0.789	0.681	14.09
merge	✓	0.0195	0.973	0.912	0.482	0.0627	0.783	0.675	13.98
merge	✗	0.0191	0.975	0.917	0.483	0.0621	0.788	0.679	14.06
merge + relabel (ours, full)	✓	0.0186	0.974	0.918	0.520	0.0613	0.784	0.692	5.43
merge + relabel	✗	0.0182	0.977	0.923	0.521	0.0606	0.788	0.695	5.43

in the pipeline. Two findings stand out. Extending the generator in place outperforms every pipeline we compared against that attaches a segmenter outside it, on the whole object and on its parts alike, and the dual-volume packing improves whole-object fidelity over the same backbone fine-tuned on the same corpus, which suggests that packing is a better representation of the same objects rather than only a container for parts.

Limitations and future work. The extension inherits the two-volume assumption from PartPacker [44]. When the contact graph is far from bipartite, parts must be merged until it two-colors, which undersegments objects whose parts interlock densely, and part quality depends on the per-part annotations the corpus provides. Like any generator, KaiNinja is stochastic, so the decomposition varies with the seed. Three failure modes recur (Figure 10). First, the layout flow sometimes undersegments. It assigns nearly the whole object to one volume and only a few small parts to the other, and parts that touch inside the crowded volume are fused, because nothing within a single volume tells them apart. Second, it sometimes collapses entirely onto one volume and leaves the other empty, so the object comes out whole with nothing to separate. Third, it sometimes does the opposite and generates the whole object in both volumes at once, so the assembled result carries two nearly coincident copies that the part metrics read as many small pieces. Although a post-processing step between the two stages merges overlapping occupancy, thin shells sometimes escape

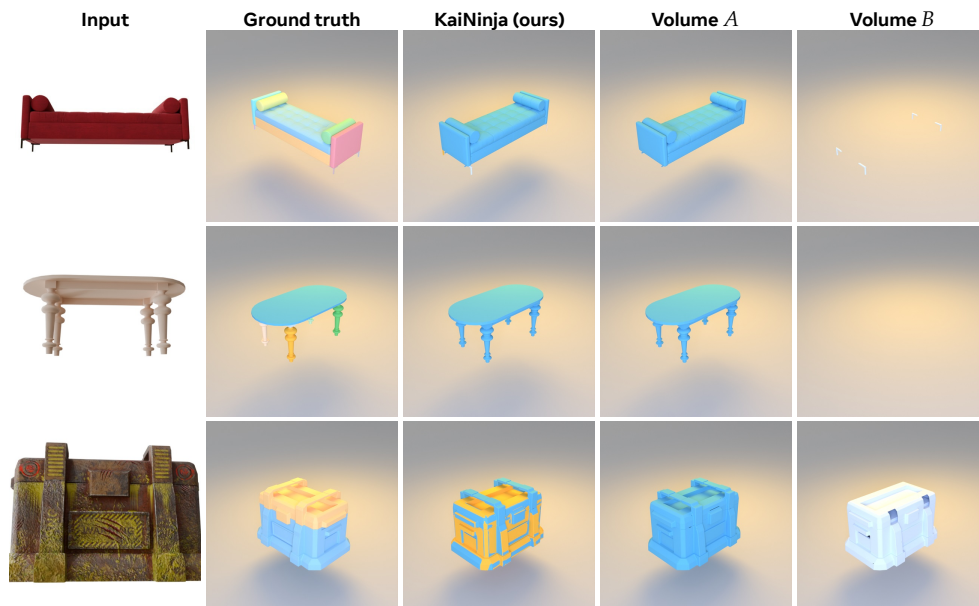


Figure 10 Three failure modes, with each volume shown on its own. *Top: undersegmentation.* The whole body of the sofa lands in volume *A* and only the four feet in volume *B*, so base, cushion, arms and bolsters, which touch and share a volume, come out as one piece. *Middle: single-volume collapse.* Volume *B* receives no surface at all; the geometry is right but the table comes out whole. This happens on 9 of the 1000 test objects. *Bottom: duplicated volumes.* Both streams generate the whole crate, with 63% of each volume’s surface within two fine voxels of the other’s, so the assembled object holds two nearly coincident copies that the part metrics read as many small pieces. Thin shells like this one sometimes escape the merge step between the two stages; the disjointness penalty alleviates this mode but does not resolve it.

the filter and lead to this failure mode; the disjointness penalty alleviates it but does not resolve it. We read all three as the limit of inheritance. The pretrained weights were shaped by undivided objects, and nothing in them asks for a balanced split or discourages the two streams from converging on the same surface. Training on packed volumes from the start would test this reading, but the compute and data it would take put it beyond this work. Separately, the Stage-2 decoder drops facets now and then and leaves small holes, so the released pipeline still runs a light geometric cleanup before delivery. Natural next steps are packing into more than two volumes, exposing explicit control over part count and granularity, learning to reassemble the separated parts, and applying the same inheritance recipe to future whole-object backbones.

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Appendix

A Hyperparameters

Both flows use a width of 1536, 30 blocks, 12 heads and rotary position encoding. They are trained in bfloat16 with AdamW [34] ($\beta = (0.9, 0.95)$, weight decay 0.01), an EMA rate of 0.9999, adaptive gradient clipping at norm 1.0, a logit-normal timestep schedule [9], and $p_{\text{uncond}} = 0.1$ for classifier-free guidance. The global batch size is 32 throughout.

Stage 1, the layout flow, works on a 16^3 sparse structure latent with 8 channels per stream and places a cross-volume attention block after depths $\{6, 12, 18, 24, 29\}$. Its frozen warmup runs 10K steps at learning rate 10^{-4} with 500 warmup steps. Joint fine-tuning runs 100K steps at 5×10^{-5} with 2000 warmup steps and the disjointness penalty at $\lambda_{\text{ov}} = 5$.

Stage 2, the refinement flow, works on the 32-channel structured latent at resolution 32 and merges the attention pools at every third block, depths $\{2, 5, 8, \dots, 29\}$. Its frozen warmup runs 30K steps and full fine-tuning 100K steps, both at learning rate 3×10^{-5} . The structured latent autoencoder is the released TRELLIS.2 one and is never trained.

B Two Volumes and Graph Coloring

Two-coloring is a heuristic, not a guarantee, because it works only when the contact graph is bipartite. PartPacker suggests lifting this restriction by making the connectivity graph planar and applying the four color theorem, but that step does not work in general. A 3D contact graph need not be planar. Tietze [46] showed that any number of regions in space can be pairwise adjacent, and Besicovitch [1] showed the same for convex polyhedra, so the complete graph on any number of parts is a 3D contact graph and no planar-map argument applies. Making such a graph planar means deleting edges. An edge can be deleted in two ways. Merging its two parts is what we and PartPacker do, at the cost of a coarser decomposition. Breaking the contact instead, by moving one part away from the other, keeps both parts but changes the assembled position of the object, so the model would generate an exploded object and a dedicated network would likely be needed to predict the displacements and reassemble it; we leave this route to future work. Deleting an edge without either of these puts two touching parts back into the same volume, which defeats the packing. In fact, the chromatic number of a 3D contact graph is unbounded, so no fixed number of volumes is safe in the worst case. PartPacker and KaiNinja both merge parts until the graph is bipartite, which undersegments objects whose parts interlock densely. Therefore, packing does not give a correct coloring in general. What it does give is a volume in which no part touches another, so each part can be generated without overwriting its neighbors.